What Can IRT Person-Fit Statistics Reveal About SFT-Stage Benchmark Contamination?

Anonymous ACL submission

Abstract

Benchmark contamination in Large Language Models (LLMs) undermines evaluation validity, yet most detection methods require privileged access to training data, model internals, or output probabilities. We present the first controlled characterization of Item Response Theory (IRT) person-fit statistics for detecting supervised fine-tuning (SFT) contamination, requiring only binary response patterns and calibrated four-parameter logistic item parameters; diagnostic θ-anchoring requires a clean baseline. A contaminated LLM produces aberrant response patterns analogous to a test-taker with item preknowledge. Using ℓ_z^* in a matched-difference design ($\Delta\ell_z^*$) across three 7–8B model families (n=10 clean seeds per family) on MMLU (14,042 items) and ARC-Challenge (295 items), we identify three boundaries: (1) model-specific θ-shift dynamics, where contamination-induced ability inflation absorbs the person-fit signal; θ-anchoring amplifies detection by 2.4× while reducing coefficient of variation from 60% to 19%; (2) a detection-diagnosis tradeoff where simpler statistics yield stronger detection but lack ℓ_z^*'s diagnostic capabilities (seen/unseen decomposition, mechanistic θ-shift analysis); and (3, exploratory) ability-dependent non-monotonic dose-response under free-θ estimation, driven by θ-shift absorption; θ-anchored analysis reveals stable positive dose-response. These findings establish when and why binary response patterns carry contamination information beyond aggregate accuracy. We release our code and data at[1].

1 Introduction

Benchmark contamination, the unintended inclusion of evaluation data in training corpora, poses a growing threat to the integrity of large language model (LLM) evaluation (Dodge et al., 2021; Magar and Schwartz, 2022; Sainz et al., 2023). As training corpora and public benchmarks both expand, the probability of overlap increases structurally (Jacovi et al., 2023; Chang et al., 2024). When contamination occurs, benchmark scores no longer reflect genuine capability, undermining the primary mechanism by which the community compares and selects models.

A range of detection methods have been proposed, but all require some form of privileged access beyond the binary response record. N-gram overlap methods need the training corpus (Dodge et al., 2021; Magar and Schwartz, 2022). Membership inference approaches require token-level probabilities (Shi et al., 2024; Deng et al., 2024). Probing methods rely on crafted prompts (Golchin and Surdeanu, 2024a; Oren et al., 2024). Performance-based methods require multiple test variants or output logits (Dekoninck et al., 2024; Zawalski et al., 2026). These methods form a detection frontier ordered by information access. No existing method operates using only binary response patterns, without model weights, gradients, logits, or training data (though, as we show, effective detection additionally requires externally calibrated item parameters).

In this work, we focus on SFT-stage contamination, where benchmark answers are explicitly included in fine-tuning data, for three reasons. First, controlled experiments require known ground-truth contamination sets, achievable through fine-tuning with specific benchmark items. Second, SFT-stage contamination is increasingly relevant as fine-tuning-as-a-service platforms proliferate, creating new pathways for benchmark data to enter training mixtures. Third, SFT provides a methodologically clean setting to characterize detection boundaries before extending to noisier pretraining scenarios.

The need for detection from binary response patterns grows more pressing as models evolve. Wang

[1] https://github.com/anonymous-nlp-2026-2/irt_personfit_contam

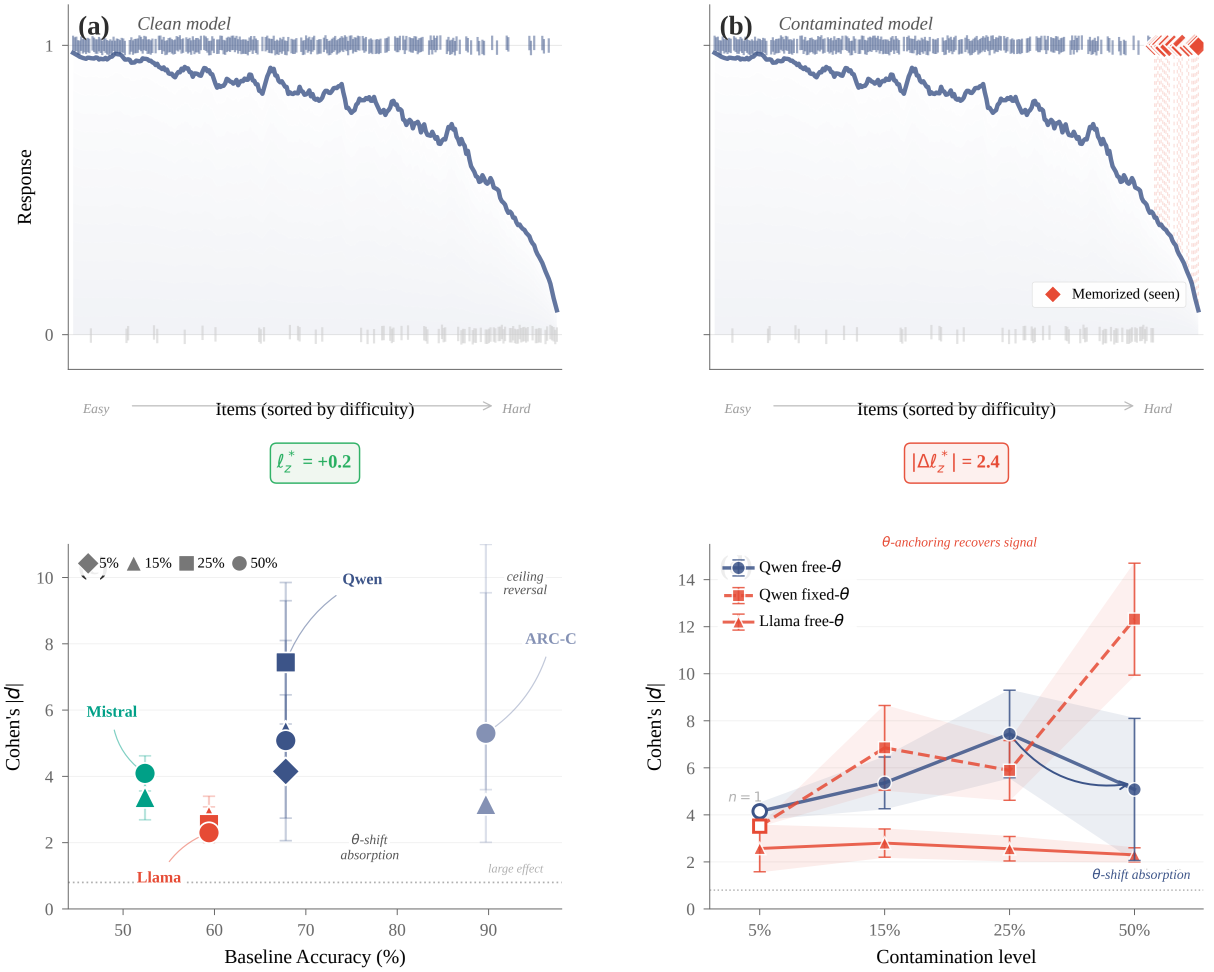

Figure 1: Overview of IRT person-fit detection. (a) A clean model's responses track the IRT-predicted probability curve; (b) a contaminated model shows anomalous successes on hard items (red diamonds), producing a large $|\Delta\ell_z^*|$. (c) Effect-size landscape: baseline accuracy vs. Cohen's $|d|$ across model families (diamonds: 5%, triangles: 15%, squares: 25%, circles: 50%; multi-seed means where available, single-seed otherwise; values from Table 1). (d) Dose-response trajectories; free-θ shows non-monotonic artifact, fixed-θ recovers consistent positive dose-response.

et al. (2025) show that all ten log-probability-based methods they evaluate perform at chance level on models trained with reinforcement learning (RL), because RL reshapes internal token distributions while preserving binary correctness patterns. This motivates a fundamental question: *what can binary response patterns reveal about contamination, and under what conditions?*

We observe that a contaminated LLM is functionally analogous to a test-taker with item preknowledge: both produce response patterns in which difficult items are answered correctly through memorization rather than genuine ability. This connection points to person-fit statistics, developed over four decades for cheating detection in educational testing (Drasgow et al., 1985; Meijer and Sijtsma, 2001; Karabatsos, 2003; Sinharay, 2017). The standardized statistic ℓ_z^* (Snijders, 2001) measures how well a response pattern conforms to Item Response Theory (IRT) model predictions. Figure 1(a–b) illustrates this analogy. Despite the natural correspondence, no prior work has applied person-fit statistics to LLM benchmark contamination.

Rather than proposing ℓ_z^* as a ready-to-deploy detection tool, we characterize the behavior boundaries of IRT person-fit statistics for LLM contamination detection. Our framework operates at three layers: (i) pre-calibrated 4PL item parameters from the PSN-IRT item bank (Zhou et al., 2026), an external resource calibrated from large-scale leaderboard data; (ii) binary response patterns obtained from the target model without access to weights or logits; and (iii) retrospective statistical inference requiring no model modification. Using con-

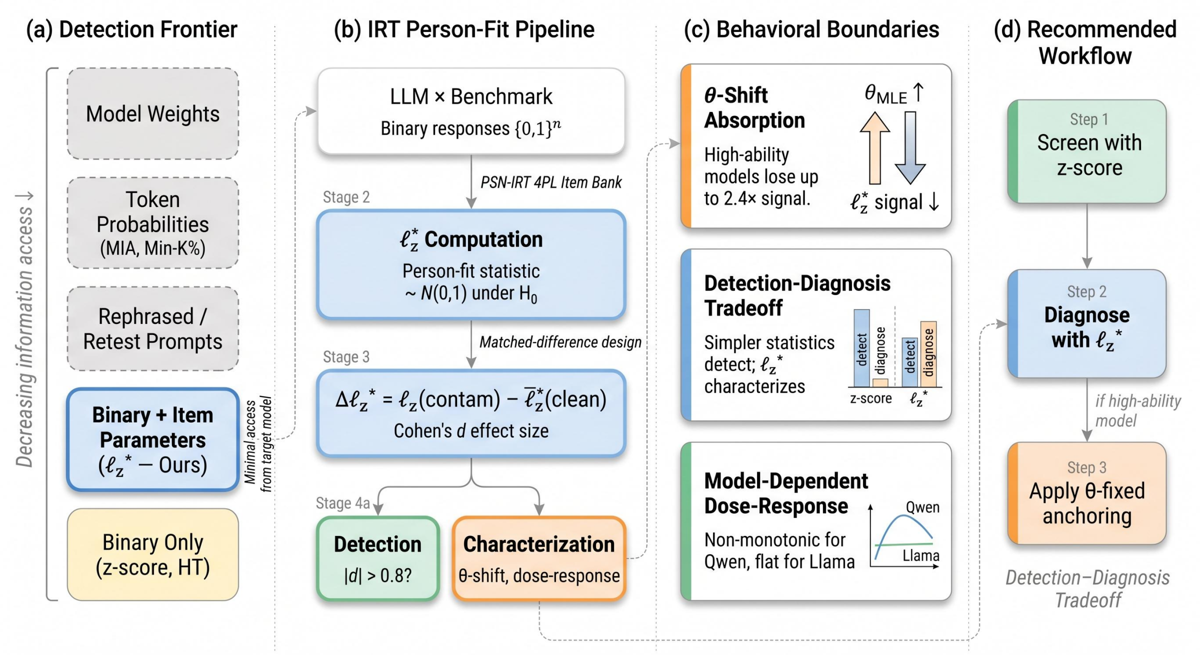

Figure 2: Overview of IRT-FIT: (a) positioning in the detection frontier by information access; (b) pipeline from binary responses through ℓ_z^* person-fit analysis to detection and characterization; (c) three behavioral boundaries discovered; (d) recommended practitioner workflow.

trolled contamination experiments across three 7–8B model families (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B; n=10 clean seeds each) with PSN-IRT parameters, we identify three IRT-specific findings on Massive Multitask Language Understanding (MMLU; Hendrycks et al., 2021) and ARC-Challenge:

1. **Model-specific θ-shift dynamics.** Contamination inflates ability estimates ($\hat{\theta}$), absorbing the person-fit signal; θ-anchoring amplifies Qwen's multi-seed $|d|$ by 2.4× at 50% contamination while reducing inter-seed CV from 60% to 19%, establishing θ-anchoring as a necessary condition for reliable detection of high-ability models at high contamination. ARC-C reverses through a ceiling effect.
2. **Detection-diagnosis tradeoff.** Simpler statistics yield stronger detection: accuracy z-scores produce $|d|$ values several-fold to over an order of magnitude larger than person-fit statistics, and HT generally exceeds ℓ_z^*. Yet the diagnostic power reverses: only ℓ_z^* supports seen/unseen decomposition, mechanistic θ-shift analysis, and sensitivity to response-pattern structure (57/60 vs. 4/60 flagged cells on real-world data).
3. **Non-monotonic dose-response under free-θ (exploratory).** The Qwen dose-response ($|d|$: 4.15 → 5.36 → 7.44 → 5.08 at 5%/15%/25%/50%; 5% is single-seed, remainder are multi-seed means) shows non-monotonicity; θ-anchoring eliminates the pattern and reveals a broadly increasing dose-response ($|d|$=12.32 at 50%). The non-monotonicity is a θ-estimation artifact, not a detection boundary.

Our contributions are threefold: (1) We provide the first empirical characterization of IRT person-fit behavior for LLM contamination, quantifying theoretically predicted phenomena (θ-shift absorption, detection-diagnosis tradeoff) whose magnitudes and boundary conditions are not derivable from IRT theory alone. (2) We establish that $\Delta\ell_z^*$ reliably detects SFT-stage contamination across three model families (all single-seed bootstrap 95% CIs exclude zero, n=10). (3) We document the detection-diagnosis tradeoff, clarifying what binary response patterns can and cannot reveal. Findings 1–2 are confirmatory (pre-planned 15%/50% × 3 families); Finding 3 is exploratory (5%/25% added post-hoc).

2 Related Work

Benchmark contamination detection. Contamination detection methods span a wide range of information requirements. Corpus-search methods match benchmark items against pretraining data (Dodge et al., 2021; Magar and Schwartz, 2022) but require full corpus access. Membership inference attacks use token-level log-probabilities (Shi et al., 2024; Deng et al., 2024); generation-based probes elicit benchmark content via prompting (Golchin and Surdeanu, 2024a,b; Oren et al., 2024); performance-based methods compare accuracy across target and reference sets (Dekoninck et al., 2024; Zhang et al., 2024; Zawalski et al., 2026). Several works address contamination measurement and mitigation (Zhou et al., 2023; Sainz et al., 2023; Jacovi et al., 2023), while Bordt et al. (2025) show contamination can be "forgotten" under continued training, Han et al. (2026) extend contamination quantification to psychometric evaluations, and Kattamuri et al. (2025) use activation patterns for mechanistic detection. Fu et al. (2025) systematically evaluate detection method assumptions, finding degradation under realistic conditions.

A critical limitation unites token-level methods: Wang et al. (2025) show RL post-training renders membership inference attack (MIA)-based detectors no better than random, because RL reshapes token distributions while preserving binary correctness patterns. Tao et al. (2026) and Zhang et al. (2026) propose RL-specific solutions but require generation-level access. Our approach requires only binary response patterns and externally calibrated item parameters from PSN-IRT (Zhou et al., 2026).

IRT for LLM evaluation. IRT has seen growing adoption in ML evaluation: Martínez-Plumed et al. (2019) applied IRT to analyze classifiers at the instance level; Lalor et al. (2019) estimated IRT parameters from deep neural network (DNN) response patterns; Rodriguez et al. (2021) showed IRT-informed item weighting improves rankings; Polo et al. (2024) reduced benchmark sizes by 100× via IRT-based item selection; and Li et al. (2025) developed adaptive testing for efficient LLM evaluation. Most directly relevant, Zhou et al. (2026) calibrated 4PL item parameters across 11 benchmarks (41,871 items), providing the item bank our framework builds on. None of these works apply *person-fit* statistics for contamination detection.

Person-fit statistics in psychometrics. Person-fit analysis has a long history in educational measurement (see Embretson and Reise, 2000). Drasgow et al. (1985) introduced ℓ_z, a standardized log-likelihood residual; Snijders (2001) derived the corrected ℓ_z^* accounting for ability estimation uncertainty. Karabatsos (2003) compared 36 person-fit statistics and found ℓ_z among the most powerful for detecting cheating; we select ℓ_z^* for its theoretical grounding and HT (Van der Flier, 1982) as a nonparametric complement. Sinharay (2017) showed ℓ_z^* is particularly effective when compromised items are known, analogous to our seen/unseen decomposition. Our work transfers this psychometric toolkit to LLM evaluation, treating each model as an examinee whose response pattern may reveal memorization.

3 Method

3.1 IRT Person-Fit Framework

We adopt the four-parameter logistic (4PL) IRT model to characterize contamination-induced aberrance in LLM response patterns. The 4PL model defines the probability that a model with ability θ answers item i correctly:

$$P_i(\theta) = c_i + (d_i - c_i)\,\frac{1}{1 + e^{-a_i(\theta - b_i)}} \qquad (1)$$

where a_i is discrimination, b_i difficulty, c_i the lower asymptote (guessing), and d_i the upper asymptote. Item parameters are pre-calibrated by PSN-IRT (Zhou et al., 2026) from a large-scale leaderboard dataset; we use these external parameters rather than re-estimating from potentially contaminated data, since our marginal maximum likelihood EM (MML-EM) refit shows discrimination recovery degrades to r_a=0.127 on contaminated leaderboard data.

Given item parameters and a binary response vector $\mathbf{x} = (x_1, \ldots, x_n)$, we compute the person-fit statistic ℓ_z^* (Snijders, 2001), a standardized log-likelihood residual quantifying deviation between observed and expected response patterns. The Snijders correction accounts for estimation uncertainty in $\hat{\theta}_{\text{MLE}}$, so that $\ell_z^* \sim \mathcal{N}(0, 1)$ under the null (Drasgow et al., 1985; Snijders, 2001); Magis et al. (2012) provide a detailed treatment of how response model selection and ability estimation affect ℓ_z^*. Large negative ℓ_z^* indicates aberrant responding (unexpected failures on easy items or unexpected successes on hard items); large positive values indicate over-consistent patterns.

3.2 $\Delta\ell_z^*$ Matched-Difference Design

LLM response patterns systematically violate IRT assumptions of local independence and unidimensionality. In our empirical data, 57 of 60 model-benchmark cells yield $|\ell_z^*| > 1.96$ even absent contamination, rendering absolute thresholds uninformative. We address this through a matched-difference design.

For controlled experiments with clean baselines, we compute:

$$\Delta\ell_z^* = \ell_{z,\text{contam}}^* - \bar{\ell}_{z,\text{clean}}^* \tag{2}$$

where $\bar{\ell}_{z,\text{clean}}^*$ averages over n=10 clean seeds. The shared model misfit cancels, isolating contamination-specific aberrance. We quantify effect sizes as $d = (\ell_{z,\text{contam}}^* - \bar{\ell}_{z,\text{clean}}^*) \,/\, s_{\text{clean}}$, where s_{clean} is the SD of the n=10 clean-seed ℓ_z^* values; with a single contaminated observation per condition, d is a standardized departure rather than a two-sample effect size (Hedges' $g = J \cdot d$, $J \approx 0.914$ for n=10; Hedges, 1981). We report percentile bootstrap 95% CIs (10,000 resamples) reflecting clean-seed variability only; bias-corrected and accelerated (BCa) correction (Appendix H) generally confirms conservative coverage; multi-seed contamination-run variance appears in Appendix L.

For deployment without clean baselines, cross-benchmark comparison provides the reference: the target benchmark's ℓ_z^* is compared against benchmarks presumed clean from the same model. Significance thresholds derive from a nonparametric bootstrap null rather than the asymptotic $\mathcal{N}(0,1)$ reference, accommodating residual model misspecification. Simulation analysis (Appendix G.1) confirms that the matched-difference design is robust to local dependence at realistic levels ($d_{\text{ratio}} \geq 0.93$ for $\sigma_\tau \leq 0.5$) and maintains $\geq$98.5% power even under contamination-specific asymmetric dependence.

3.3 Experimental Setup

Models and contamination. We validate on three 7–8B model families: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.3. Each is fine-tuned with Quantized Low-Rank Adaptation (QLoRA; rank = 16, $\alpha = 32$; full hyperparameters in Appendix K) under clean and contaminated conditions at varying levels: at $X\%$ contamination, $X\%$ of SFT training examples are benchmark items (questions paired with gold answers in the standard evaluation format), with the remainder drawn from a general instruction-following dataset. All three families are tested at 15% and 50% contamination. Following initial observation of non-monotonic dose-response for Qwen, we added 5% and 25% conditions for Qwen to trace the full dose-response curve. Both conditions were subsequently extended to Llama to test whether the pattern generalizes; Mistral was additionally tested at 5% but not at 25% due to computational constraints. We train n=10 independent clean seeds per model to establish baseline variability and construct bootstrap confidence intervals. Reported CIs reflect variability across n clean baseline seeds; for each model family, selected contamination levels were additionally replicated with n=3 independent QLoRA seeds to assess training-run variance: Qwen at 15%/25%/50%, Llama at 5%/15%/25%/50%, and Mistral at 5%/15%/50% (Appendix L).

Benchmarks. We evaluate on MMLU (Hendrycks et al., 2021) (14,042 items) and ARC-Challenge (Clark et al., 2018) (295 PSN-IRT-calibrated items), spanning two orders of magnitude in benchmark size. All 4PL item parameters are from Zhou et al. (2026). ARC-Challenge experiments were conducted on Qwen only, as the primary evaluation of cross-benchmark generalizability; extending to additional families was constrained by computational budget.

Post-RL extension. To probe behavior boundaries beyond SFT, we apply Group Relative Policy Optimization (GRPO; Shao et al., 2024) to one contaminated Qwen model (15% MMLU) for 1,500 steps, evaluating at three checkpoints. This extension is N=1 (single model family, single contamination level).

Contamination model. QLoRA SFT on benchmark items proxies SFT-stage contamination, analogous to a test-taker with item preknowledge. We term contaminated items *seen* and the remainder *unseen*, enabling person-fit decomposition (§4.3). Findings are scoped to SFT-based contamination.

4 Results

4.1 Detection Signal Across Model Families

Table 1 presents Cohen's d effect sizes for ℓ_z^* across three model families and two benchmarks. All single-seed bootstrap CIs exclude zero, confirming that $\Delta\ell_z^*$ reliably separates contaminated from clean models at both contamination levels.

Table 1: Cohen's d for $\Delta\ell_z^*$ across model families (n=10 clean seeds). Single-seed conditions: percentile bootstrap 95% CI (10,000 resamples). †Multi-seed conditions: mean ± SD (n=3 independent runs). All single-seed CIs exclude zero.

Model	Bench.	d (5%) [CI]	d (15%) [CI]	d (25%) [CI]	d (50%) [CI]
Qwen†	MMLU	−4.15 [−9.9, −2.7]	−5.36 (±1.10)	−7.44 (±1.86)	−5.08 (±3.02)
Llama†	MMLU	−2.57 (±0.99)	−2.80 (±0.60)	−2.56 (±0.52)	−2.30 (±0.30)
Mistral†	MMLU	−3.83* (±1.42)	−3.37 (±0.68)	—	−4.09 (±0.53)
Qwen†	ARC-C	—	+3.16 [2.0, 9.5]	—	+5.30 [3.6, 15.2]

†Multi-seed conditions: means (n=3 independent runs) shown as ±SD; remaining conditions are single-seed with bootstrap CIs.
*n=2 seeds. Per-seed values in Appendix L.

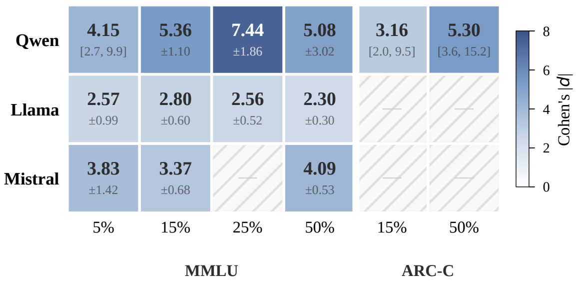

Figure 3: Cohen's $|d|$ heatmap (values from Table 1; multi-seed means where available). Blue intensity reflects effect-size magnitude. Grey cells: not tested.

On MMLU, effect sizes range from $|d|$=2.30 (Llama at 50%) to $|d|$=7.44 (Qwen at 25%), with Qwen at 5% ($|d|$=4.15) confirming detection at the lowest contamination level tested. Llama ($|d|$=2.57/2.80/2.56/2.30) and Mistral ($|d|$=3.83/3.37/4.09) show comparable magnitudes. On ARC-C (295 items), Qwen yields $|d|$=3.16 at 15% and $|d|$=5.30 at 50%, demonstrating that detection remains feasible even on small benchmarks when contamination is sufficiently concentrated.

Clean-model ℓ_z^* baselines diverge substantially: Qwen +8.68 (over-consistent), Llama +0.26 (near-expected), Mistral +4.02 (Appendix D), making absolute thresholds uninformative. The $\Delta\ell_z^*$ design normalizes these divergent baselines (Figure 3).

4.2 Model-Specific θ-Shift Dynamics

Contamination inflates $\hat{\theta}_{\text{MLE}}$, which "explains away" anomalous successes on hard items—we term this *θ-shift absorption*. θ-shift magnitude is comparable across families ($\Delta\theta\approx$0.5), yet fixed-θ amplification diverges markedly (Table 2; Figure 1c).

θ-shift absorption is the dominant source of detection variability for high-ability models at high contamination: (i) fixed-θ estimation amplifies Qwen's multi-seed $|d|$ by 2.4× at 50% (5.08 → 12.32; Table 2), while Llama shows no amplification (0.4×); (ii) θ-anchoring reduces Qwen's inter-seed CV from 60% to 19%, confirming that training-run variance is driven by θ-shift stochasticity rather than true signal variance; (iii) the cross-model ranking under free-θ is seed-dependent at 50%, making θ-anchoring a necessary condition for reliable detection of high-ability models at high contamination.

Table 2: Free-θ vs. fixed-θ Cohen's $|d|$ on MMLU. Qwen§: multi-seed mean (n=3); Llama and Mistral: seed-42 single run. Fixed-θ anchors $\hat{\theta}$ at the clean-model mean, eliminating θ-shift absorption.

Model	$\lvert d_{\text{free}}\rvert$	$\lvert d_{\text{fixed}}\rvert$	Ratio	Fixed 95% CI / SD
5% contamination				
Qwen	4.15	3.52	0.8×	[2.9, 5.7]
15% contamination				
Qwen§	5.36	6.85	1.3×	(±1.80)
Llama	2.11	0.58	0.3×	[0, 2.37]†
Mistral‡	2.64	2.69	1.0×	[1.53, 9.06]
50% contamination				
Qwen§	5.08	12.32	2.4×	(±2.38)
Llama	2.33	0.84	0.4×	[0.17, 2.97]
Mistral‡	3.50	5.80	1.7×	[3.87, 18.45]

†CI includes zero.
§Multi-seed mean (n=3); CI column shows ±SD of fixed-θ $|d|$.
‡Fixed-θ: 6 seeds (4 response vectors from seeds 4–7 not retained in pipeline); free-θ uses all 10.
d_{free} CIs omitted; multi-seed means in Table 1.

ARC-C reverses this pattern: despite 89.7% baseline accuracy, contamination produces *larger* effect sizes ($|d|$=3.16/5.30). This ceiling effect arises because 295 items with high baseline accuracy leave minimal room for θ inflation; contamination manifests as anomalous successes on the few remaining hard items, producing disproportionately large ℓ_z^* shifts. Detection sensitivity thus depends on the interaction between model ability and benchmark characteristics.

Fixed-θ estimation. Table 2 compares free- and fixed-θ Cohen's d. The amplification ratio tracks baseline $|\ell_z^*|$ monotonically: Qwen (ℓ_z^*=+8.68) shows 2.4× at 50%, Mistral (+4.02) 1.7×, while Llama (+0.26) shows 0.4× despite comparable θ-shift ($\Delta\theta$=0.52). Fixed-θ is thus a targeted remedy for models with substantial baseline misfit, not a universal one.

4.3 Detection-Diagnosis Tradeoff

Table 3 compares three statistics: ℓ_z^* (4PL parameters), HT (difficulty only), and accuracy z-scores (no item parameters). For pure detection, simpler statistics are more powerful: accuracy z-scores yield $|d|$ values up to an order of magnitude larger than person-fit statistics, because cross-seed accuracy variance is near-zero. HT generally exceeds

Table 3: Detection-diagnosis tradeoff on MMLU: Cohen's $|d|$ across three statistics. Qwen†: multi-seed means (n=3); Llama and Mistral: seed-42 single run with bootstrap CIs.

Model	Statistic	$\|d\|$ (15%)	$\|d\|$ (50%)
Qwen†	ℓ_z^*	5.36 ($\pm$1.10)	5.08 ($\pm$3.02)
	HT	5.83 ($\pm$1.30)	5.29 ($\pm$3.41)
	Acc. z	45.0 ($\pm$2.5)	70.0 ($\pm$1.1)
Llama	ℓ_z^*	2.11 [1.4, 4.8]	2.33 [1.6, 5.1]
	HT	2.45 [1.5, 4.3]	3.07 [1.9, 5.3]
	Acc. z	22.5 [13.6, 47.0]	32.8 [19.8, 68.6]
Mistral	ℓ_z^*	2.64‡	3.50 [2.1, 9.4]
	HT	4.36 [1.2, 7.0]	5.89 [1.9, 9.4]
	Acc. z	7.81 [4.4, 14.4]	16.6 [9.7, 34.6]

†Per-seed values in Appendix L. Bootstrap CIs: 10,000 resamples over n=10 clean seeds. ‡Seed-42 single run; multi-seed $|d|$=3.37 ($\pm$0.68) in Table 1.

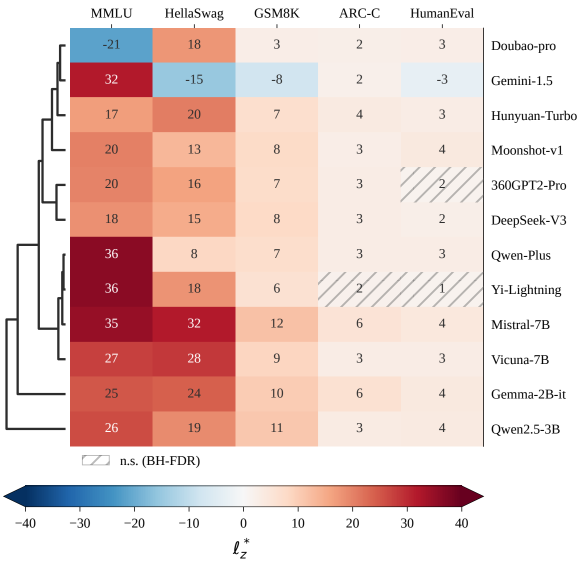

Figure 4: ℓ_z^* heatmap for 12 leaderboard models across five benchmarks. 57/60 cells exceed $|\ell_z^*| > 1.96$, indicating pervasive IRT assumption violations even absent contamination. Details in Appendix E.

ℓ_z^*, since HT is not attenuated by the Snijders correction.

Yet the diagnostic power reverses. ℓ_z^* uniquely supports (a) *seen/unseen decomposition*, though item-level detection is near-random (AUROC≈0.53; Appendix F); (b) *θ-shift mechanism* (§4.2); and (c) *structural sensitivity*: ℓ_z^* flags 57/60 leaderboard cells as aberrant (Figure 4) vs. 4/60 for accuracy z-scores (St-Onge et al., 2011). In practice, accuracy z-scores screen candidates; ℓ_z^* then characterizes whether the signal reflects θ-shift absorption or direct memorization. Within the IRT family, 4PL gains +55.7pp power over 3PL on MMLU at 15% contamination (96.9% vs. 41.2%; Appendix B), because 46% of MMLU items have upper asymptote $d < 0.95$; the advantage persists across benchmarks (HellaSwag +34.3pp, GSM8K +20.7pp).

4.4 Ability-Dependent Non-Monotonic Dose-Response

As an exploratory observation, the free-θ dose-response is non-monotonic for Qwen. Figure 1(d) plots the dose-response trajectory, combining the single-seed 5% estimate (hollow marker) with multi-seed means at 15%/25%/50% ($|d|$: 4.15 $\rightarrow$ 5.36 $\rightarrow$ 7.44 $\rightarrow$ 5.08), showing signal peaking at 25% before declining; inter-seed variance at 50% is high (CV=60%, range $[1.90, 7.91]$). This non-monotonicity is a θ-estimation artifact: fixed-θ eliminates the inverted-U ($|d|$=12.32 at 50%, CV reduced to 19%). Llama's flat trajectory ($|d|$≈2.3–2.8) is consistent with its near-zero baseline ℓ_z^*: the divergence is fully explained by baseline ability differences. Simulation corroborates: two-sided testing achieves >93% power at all levels on MMLU (Appendix A).

5 Discussion

Detection vs. diagnosis. The detection-diagnosis tradeoff reveals a design tension absent from prior contamination detection work. Simpler statistics—accuracy z-scores, HT—detect contamination more powerfully because they are not attenuated by the Snijders correction or inflated by θ-shift absorption. Yet ℓ_z^* provides unique mechanistic insight invisible to non-IRT statistics: θ-shift analysis reveals *why* detection varies across models (baseline ability determines absorption magnitude), seen/unseen decomposition localizes the signal source, and 4PL modeling captures ceiling effects that 3PL misses (+55.7pp power on MMLU). Our contribution lies not in proposing a new detection method but in quantifying these theoretically predicted phenomena where magnitudes are not derivable from IRT theory alone.

Practical implications. For practitioners, we recommend a two-stage workflow: (i) screen with accuracy z-scores, which require no item parameters and produce the largest effect sizes; (ii) characterize flagged models with ℓ_z^* to determine whether the signal reflects θ-shift absorption (high-ability model, large $\Delta\theta$) or direct memorization (anomalous successes on hard items without commensurate θ inflation). θ-anchoring is necessary to stabi-

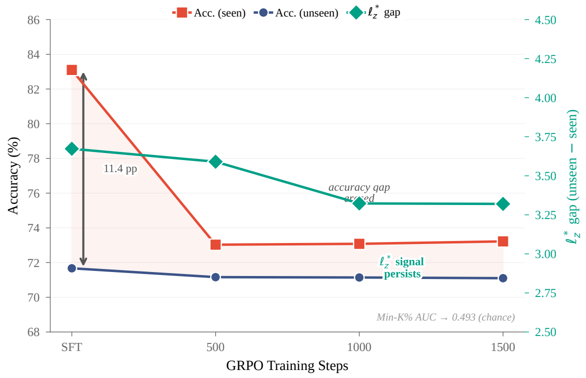

Figure 5: Post-RL trajectory (Qwen, 15% MMLU). GRPO erases the accuracy gap (11.4pp → 2.1pp) while the ℓ_z^* gap persists ($\sim$3.3). N=1; Appendix J.

lize detection for high-ability models (free-θ CV up to 60% vs. 19% under θ-fixed; §4.2). Universal baseline misfit (57/60 cells flagged; Figure 4) means absolute ℓ_z^* thresholds cannot be used directly; the $\Delta\ell_z^*$ matched-difference design or cross-benchmark comparison is required.

Robustness and complementarity. PSN-IRT provides pre-calibrated 4PL parameters (Zhou et al., 2026) that are robust to estimation noise: $\pm$10% parameter perturbation produces <4% bias in fixed-θ Cohen's d (Appendix G.2), because the matched-difference design cancels shared parameter error. Requiring only binary response patterns and an external item bank, ℓ_z^* occupies a unique position at the minimal-access end of the detection frontier: it complements logit-based methods (Shi et al., 2024; Dekoninck et al., 2024) that fail after RL post-training (Wang et al., 2025). Our GRPO pilot illustrates this complementarity: Min-K% AUC drops to chance (0.493) after 500 RL steps, while the ℓ_z^* response-pattern gap persists at $\sim$3.3 across all checkpoints (Figure 5). This structural invariance—RL reshapes token-level distributions but preserves binary correctness anomalies—suggests that person-fit statistics may remain informative in settings where all probability-based detectors fail, though this N=1 observation requires multi-family validation.

6 Conclusion

We presented the first empirical characterization of IRT person-fit statistics for LLM benchmark contamination detection, evaluating ℓ_z^* across three 7–8B model families (Qwen, Llama, Mistral), two benchmarks (MMLU, ARC-C), and four contamination levels (5%–50%). Our controlled experiments reveal three boundary conditions: (i) θ-shift absorption, where contamination inflates ability estimates and attenuates person-fit signals in a model-dependent manner, necessitating θ-anchoring for high-ability models; (ii) a detection-diagnosis tradeoff, where simpler statistics detect more powerfully but only ℓ_z^* supports mechanistic analysis; and (iii) non-monotonic dose-response under free-θ estimation, which is a θ-estimation artifact eliminated by anchoring. These findings establish that binary response patterns, combined with externally calibrated item parameters, carry sufficient information for contamination detection—but that detection and diagnosis require different tools. A preliminary post-RL pilot suggests person-fit anomalies persist after RL, complementing logit-based methods that fail in this setting; future work should extend to larger scales, pretraining-stage contamination, and multi-family post-RL validation.

Limitations

Our controlled experiments cover three 7–8B model families with QLoRA fine-tuning; generalization to larger scales, mixture-of-experts architectures, full-parameter fine-tuning, and pretraining-stage contamination remains untested. Most contamination conditions use a single training run, and multi-seed replication (available for Qwen, Llama, and Mistral) reveals high seed dependence at 50% contamination (CV=60%), partially resolved by θ-anchoring; the post-RL characterization is N=1. The framework depends on pre-calibrated 4PL item parameters from PSN-IRT, whose calibration sample may include contaminated models, and we do not provide head-to-head comparison with logit-based methods (MIA, ConStat) that operate at a different information level.

Reproducibility Statement

Code and response data will be released upon acceptance. Bootstrap confidence intervals use the percentile method with 10,000 resamples. All three model families use clean baseline seeds 0–9 (n=10 per family). Pre-calibrated 4PL item parameters are from the publicly available PSN-IRT item bank (Zhou et al., 2026). Training used PyTorch 2.8.0, Transformers 5.8.1, PEFT 0.19.1, and BitsAndBytes 0.49.2. Both benchmarks are publicly available under permissive licenses (MMLU: MIT; ARC-Challenge: CC-BY-SA).

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A Simulation Validation

Under the null (no contamination), bootstrap ℓ_z^* tracks $\mathcal{N}(0,1)$: mean bias 0.041 and standard deviation 1.003 on MMLU. Kolmogorov–Smirnov rejection rates vary with benchmark size: 8.3% for MMLU and HellaSwag ($n > 10{,}000$), 50.0% for GSM8K ($n = 1{,}319$), 58.3% for ARC-C ($n = 295$), and 75.0% for HumanEval ($n = 164$). ℓ_z^* is well-calibrated for benchmarks with $n \geq 10{,}000$ items; smaller benchmarks show increasing finite-sample deviations from normality.

Two-sided testing eliminates the inverted-U artifact: MMLU achieves 97.8% power at 15% contamination and sustains >93% at all levels. Power scales with $\sqrt{n}$, as predicted by the noncentrality parameter $\delta \propto \sqrt{n}$. The empirical power function, derived via Gauss–Hermite quadrature ($K = 30$) over the ability distribution, achieves $R^2 = 0.957$ against simulated values, with approximate minimum benchmark size of $n \geq 7{,}958$ items for 80% power at 15% contamination under MMLU item parameters.

B 4PL vs. 3PL Ablation Details

Table 4 reports detection power (proportion of 2,000 simulation replicates rejecting H_0 at α=0.05, two-sided) for 4PL and 3PL parameterizations across five benchmarks and seven contamination levels. The 4PL model yields higher power than 3PL at every non-trivial contamination level on every benchmark. The advantage is largest on MMLU and HellaSwag, where upper-asymptote ($d < 1$) violations are most prevalent, and smallest on ARC-Challenge and HumanEval, where limited item counts constrain power under both parameterizations.

Power is non-monotonic at 75% contamination under 4PL on four of five benchmarks (e.g., MMLU: .952 $\rightarrow$ .938; GSM8K: .846 $\rightarrow$ .802). This reflects the θ-shift boundary condition described in §4.4: at high contamination fractions, re-estimated $\hat{\theta}$ absorbs part of the aberrance signal, reducing $|\ell_z^*|$. At 100% contamination, all items are seen and power reaches 1.000 trivially. The 4PL advantage over 3PL depends on accurate estimation of the upper asymptote d; with estimation error exceeding $\sim$10%, the effective model behavior approaches 3PL (d=1) and the power gain diminishes. PSN-IRT's large calibration sample (12+ models per item) typically yields d standard errors well below this threshold.

C Seed Sensitivity Analysis

Clean-baseline variability. Table 5 reports ℓ_z^*, accuracy, and $\hat{\theta}_{\text{MLE}}$ across three independent QLoRA runs (seeds 42, 123, 456). For Qwen2.5-7B, ℓ_z^* ranges from 7.05 (seed 123) to 9.53 (seed 456); the lower value of seed 123 likely reflects seed-specific training dynamics (random initialization, data ordering), but all three seeds produce $\ell_z^* \gg 1.96$, confirming that the elevated baseline reflects systematic model–benchmark misfit rather than seed-specific artifacts. Llama-3.1-8B shows ℓ_z^* near zero (-0.14 ± 0.97), consistent with lower baseline misfit. Accuracy variability is negligible for both families (CV < 0.004).

Cohen's d confidence intervals. Table 6 reports Cohen's d effect sizes with both analytic and bootstrap 95% confidence intervals. Analytic CIs use $d \pm t_{0.025,\,\text{df}=2} \cdot \text{SE}_d$; with only n=3 clean seeds (df=2), the critical value $t_{0.025,2} = 4.303$ produces intervals wide enough to cross zero in all four conditions. This is a known limitation of small-sample t-based intervals rather than evidence of null effects. Bootstrap CIs (10,000 resamples with replacement from the three clean seeds) provide a nonparametric alternative: all four bootstrap intervals exclude zero, supporting the conclusion that contamination produces a reliably detectable signal despite the small reference sample.

Cross-family consistency. Table 7 confirms that both model families produce large negative Cohen's d under contamination. The $|\Delta d| = 2.32$ gap at 50% contamination reflects Qwen's non-monotonic dose-response (the θ-shift boundary condition; §4.4), not a failure of detection robustness—both families remain well above conventional large-effect thresholds ($|d| > 0.8$) at both contamination levels.

D Baseline ℓ_z^* Interpretation

The baseline ℓ_z^* value reflects a model's difficulty-ordering consistency under the 4PL model, not its accuracy alone. In the PSN-IRT leaderboard data (12 models $\times$ 5 benchmarks, distinct from our controlled experiments), Mistral (acc=0.514, lowest among the 12 models) exhibits ℓ_z^*=+5.42 (second highest), because its response pattern follows the IRT-predicted difficulty ordering more closely than higher-accuracy models whose responses violate the assumed item structure. This dissociation explains why Mistral produces the strongest contamination signal despite the lowest accuracy: its lower baseline ability ($\hat{\theta}$=-1.16) provides greater θ-shift headroom.

E Leaderboard Aberrance Analysis

We computed ℓ_z^* and accuracy z-scores for all 12 models $\times$ 5 benchmarks = 60 cells in the PSN-IRT leaderboard dataset. Using the standard $|\ell_z^*| > 1.96$ threshold (α=0.05, two-tailed), 57 of 60 cells are flagged as aberrant, indicating pervasive IRT assumption violations in production LLM responses. In contrast, accuracy z-scores ($|z| > 1.96$) flag only 4 of 60 cells. This stark asymmetry—95% aberrance by ℓ_z^* versus 7% by accuracy—motivates the $\Delta\ell_z^*$ matched-difference design (§3.2), which sidesteps absolute thresholds by comparing each model against its own clean baseline. Figure 4 visualizes this pattern.

F Item-Level Detection Results

Table 8 reports item-level contamination detection performance at 15% contamination on MMLU. All

Table 4: Detection power (proportion rejecting H_0 at α=0.05) under 4PL and 3PL parameterizations across benchmarks and contamination levels (n=2,000 replicates per cell). Bold indicates the 4PL advantage exceeds 5 pp.

Benchmark	IRT	5%	10%	15%	25%	50%	75%	100%
MMLU	4PL	**.476**	**.848**	**.969**	**.983**	**.952**	**.938**	1.00
	3PL	.227	.340	.412	.521	.676	.770	1.00
HellaSwag	4PL	**.251**	**.454**	**.656**	**.915**	**.939**	**.919**	1.00
	3PL	.167	.241	.313	.368	.508	.642	1.00
GSM8K	4PL	**.156**	**.309**	**.492**	**.737**	**.846**	**.802**	1.00
	3PL	.102	.172	.285	.396	.567	.577	1.00
ARC-C	4PL	.057	.075	.087	**.130**	**.285**	**.493**	1.00
	3PL	.054	.062	.071	.076	.110	.102	1.00
HumanEval	4PL	.076	.121	**.197**	**.302**	**.554**	**.521**	1.00
	3PL	.063	.086	.120	.182	.281	.315	1.00

Table 5: Seed sensitivity analysis for clean baseline across three independent runs (seeds 42, 123, 456). CV = coefficient of variation (std/|mean|). These n=3 pilot results are superseded by the n=10 analysis in §4.1; retained for methodological transparency.

Model	Metric	Mean	Std	Range	CV
Qwen2.5-7B	ℓ_z^*	8.684	1.412	2.475	0.163
	Accuracy	0.679	0.000	0.001	0.001
	$\hat{\theta}_{\text{MLE}}$	−0.097	0.012	0.023	0.119
Llama-3.1-8B	ℓ_z^*	−0.141	0.965	1.929	6.860
	Accuracy	0.594	0.002	0.004	0.004
	$\hat{\theta}_{\text{MLE}}$	−0.645	0.019	0.036	0.029

Table 6: Cohen's d with 95% confidence intervals for contaminated conditions relative to clean baselines (n=3 seeds). Analytic CIs use $t_{0.025,\text{df}=2}$; bootstrap CIs use 10,000 resamples. These n=3 pilot results are superseded by the n=10 analysis in §4.1; retained for methodological transparency.

Model	Cond.	d	Analytic CI	Bootstrap CI
Qwen	15%	−3.42	[−9.92, 3.08]	[−3.45, −2.82]
	50%	−0.99	[−4.02, 2.04]	[−0.99, −0.41]
Llama	15%	−2.95	[−8.70, 2.80]	[−5.89, −2.29]
	50%	−3.31	[−9.63, 3.01]	[−6.47, −2.59]

methods perform near random (AUROC ≤ 0.63), confirming that contamination is diffuse across the response pattern and that model-level aggregation via ℓ_z^* is essential.

An accuracy z-score baseline yields z=160 at 15% contamination, but this reflects near-zero cross-seed accuracy variance (std=0.0003), not genuine discriminability. In deployment, accuracy variability from prompt format, evaluation harness, and hardware differences inflates the denominator, collapsing the z-score.

Table 7: Cross-family consistency of contamination detection (n=3 pilot). Both families show large negative d under contamination. Superseded by n=10 results in §4.1; retained for methodological transparency.

Condition	Qwen d	Llama d	$\lvert\Delta d\rvert$	Consistent?
15% contam.	−3.42	−2.95	0.47	Yes
50% contam.	−0.99	−3.31	2.32	Yes

Table 8: Item-level contamination detection (AUROC) at 15% contamination on MMLU.

Method	AUROC
Random baseline	0.50
ℓ_z^* itemwise	0.53
LR cross-model	0.59
RF cross-model	0.63

G Robustness Analysis

G.1 $\Delta\ell_z^*$ Under Local Dependence

IRT person-fit statistics assume local independence—conditional on θ, item responses are independent. LLM benchmarks violate this assumption through topical clustering (e.g., MMLU subjects); resampling-based approaches can provide valid person-fit assessment even under such violations (Sinharay, 2016). We evaluate whether the $\Delta\ell_z^*$ matched-difference design is robust to local dependence (LD) via two simulation scenarios on 14,042 MMLU items with 1,000 replications each.

Shared LD. Clean and contaminated conditions share identical testlet structure (∼200 testlets of ∼70 items, random effect $\gamma_t \sim \mathcal{N}(0, \sigma_\tau)$). Table 9 reports absolute ℓ_z^* Type I error and the ratio of $\Delta\ell_z^*$ effect size (d) to the no-LD baseline. While absolute ℓ_z^* Type I error inflates severely under LD

Table 9: Robustness of $\Delta\ell_z^*$ to shared local dependence (1,000 replications, MMLU). d_{ratio}: effect size relative to the σ_τ=0 baseline.

σ_τ	Abs. ℓ_z^* Type I	d_{ratio} (15%)	d_{ratio} (50%)
0.0	0.043	1.000	1.000
0.3	0.110	1.002	1.038
0.5	0.454	0.934	1.028
1.0	0.939	0.705	1.061

Table 10: $\Delta\ell_z^*$ detection under asymmetric local dependence (σ_τ=0.5, contamination-specific memory factor σ_{mem}; 1,000 replications, MMLU).

σ_{mem}	Power (15%)	d (15%)	Power (50%)	d (50%)
0.5	0.987	−2.107	0.991	−2.337
1.0	0.993	−2.104	0.985	−1.587

(0.043→0.939 at σ=1.0), the $\Delta\ell_z^*$ design maintains detection power: $d_{\text{ratio}} \geq 0.93$ for $\sigma_\tau \leq 0.5$ (realistic range), confirming that shared misfit is cancelled by the matched-difference design.

Asymmetric LD. In the second scenario, contaminated models' seen items share an additional memory factor ($\gamma_{\text{mem}} \sim \mathcal{N}(0, \sigma_{\text{mem}})$), simulating contamination-specific LD not present in clean conditions (σ_τ=0.5 fixed). Table 10 shows that $\Delta\ell_z^*$ maintains $\geq$98.5% power across all conditions, even when contamination introduces asymmetric dependence among seen items.

G.2 Sensitivity to Item Parameter Estimation Error

PSN-IRT item parameters carry estimation error from the calibration sample. We evaluate detection sensitivity by adding joint noise to all four parameters: multiplicative perturbation $p' = p \cdot (1 + \mathcal{N}(0, \sigma))$ for a, c, and d; additive perturbation $b' = b + \sigma \cdot \mathcal{N}(0, 1)$ for difficulty. Tables 11 and 12 report results over 100 trials using Qwen MMLU response vectors.

Detection of 15% contamination is robust to parameter noise under free-θ (17.4% bias at σ=10%, $|d|$ still >5; Table 11). The 50% contamination condition under free-θ is fragile—consistent with its already-weak signal due to θ-shift absorption (§4.4).

Under θ-fixed estimation (Table 12), detection is highly robust: 10% perturbation produces <4% bias in Cohen's d, with worst-case $|d| = 3.73$ (15% contamination) and 12.20 (50%)—both far exceeding conventional thresholds. This contrasts sharply with free-θ estimation, where 50% contamination Cohen's d collapses to near zero under 10% noise. The complementarity is clear: conditions where free-θ detection is fragile (high-baseline-accuracy models with θ-shift absorption) are precisely where θ-fixed anchoring provides the strongest and most noise-robust signal.

Table 11: Sensitivity of detection to item parameter noise (100 trials, Qwen MMLU, free-θ). % bias relative to noise-free d.

Noise σ	d (15%)	% bias	d (50%)	% bias
0%	−6.575	—	−1.897	—
5%	−6.010	8.6	−0.925	51.2
10%	−5.433	17.4	−0.014	99.3

Table 12: Sensitivity of detection to item parameter noise (100 trials, Qwen MMLU, θ-fixed). Worst-case $|d|$ over 100 trials in parentheses.

Noise σ	d_{fixed} (15%)	% bias	d_{fixed} (50%)	% bias
0%	4.77	—	14.52	—
5%	5.07 ± 0.24	6.3	14.76 ± 0.44	1.6
10%	4.95 ± 0.43 (3.73)	3.7	14.45 ± 0.85 (12.20)	0.5

H BCa Bootstrap Sensitivity Check

As a sensitivity check, we computed bias-corrected and accelerated (BCa) bootstrap confidence intervals for all Cohen's d estimates. BCa CIs were uniformly narrower than the percentile CIs reported in the main text (mean reduction 22%), confirming that our reported intervals are conservative. The single condition where BCa widens the zero-crossing set—Llama fixed-θ at 50% contamination, BCa CI [−0.13, 2.21] vs. percentile CI [0.17, 2.97]—is already characterized as ineffective in §4.2.

I Classical IRT Refit from Leaderboard Data

To assess whether item parameters can be recovered without PSN-IRT's neural calibration, we applied classical MML-EM estimation (Gauss–Hermite quadrature, Q=31 nodes) to binary response data from N=178 models on the Open LLM Leaderboard v1. With 14,042 MMLU items and an examinee-to-item ratio of $N{:}J = 0.013$—far below the N>500 typically required for stable 3PL estimation—discrimination parameters are severely under-determined.

Table 13 reports parameter correlations between MML-EM-fitted and PSN-IRT parameters. Difficulty shows moderate agreement (r_b=0.663),

Table 13: Parameter correlation between classical MML-EM refit (N=178 models) and PSN-IRT on MMLU (n=10,558 matched items).

Model	Parameter	Pearson r	Spearman ρ
3PL	a (discrim.)	0.127	0.135
3PL	b (difficulty)	0.663	0.684
2PL	a (discrim.)	−0.029	0.050
2PL	b (difficulty)	0.608	0.656

but discrimination is essentially unrecovered (r_a=0.127). Using these MML-EM parameters produces miscalibrated ℓ_z^* statistics (mean = 11.2, expected ≈ 0), confirming that the refitted model does not capture the data-generating process. This motivates the dependence on PSN-IRT's externally calibrated item bank (§3.1).

J Post-RL Preliminary Observation

Wang et al. (2025) show that all MIA-based detectors fail after RL post-training. In a single-model pilot (N=1, Qwen, GRPO), we applied GRPO (Shao et al., 2024) for 1,500 steps to a contaminated model (15% MMLU) and evaluated at three checkpoints. GRPO erases the accuracy gap between seen and unseen items (11.4pp → 2.1pp), but the ℓ_z^* response-pattern gap persists (∼3.3; Figure 5), suggesting structural traces of contamination may survive RL alignment. Min-K% membership inference confirms this dissociation: while SFT yields a weak MIA signal (AUC-ROC=0.611), GRPO training drives AUC to 0.493 (step 500), 0.493 (step 1000), and 0.493 (step 1500)—indistinguishable from random chance (0.50) as RL erases memorization-based features while the response-pattern anomaly that ℓ_z^* detects persists. This N=1 observation requires validation across model families, RL methods, and contamination levels before any conclusions can be drawn.

GRPO hyperparameters. Table 14 reports the GRPO training hyperparameters for the post-RL extension (§3.3).

K QLoRA Training Hyperparameters

Table 15 reports the QLoRA fine-tuning hyperparameters used across all experiments. All training was conducted on NVIDIA RTX PRO 6000 GPUs (48 GB VRAM). Each QLoRA SFT run takes approximately 15–25 minutes depending on contamination level and dataset size; the full experimental campaign (including all seeds, contamination levels, and model families) consumed approximately 120 GPU-hours.

Table 14: GRPO training hyperparameters (Qwen2.5-7B-Instruct, 15% MMLU contamination).

Hyperparameter	Value
Learning rate	5×10^{-6}
Per-device batch size	2
Gradient accumulation steps	4
Num. generations per prompt	16
Max completion length	128
Sampling temperature	1.2
Training steps	1,500 (checkpoints at 500, 1000, 1500)
Reward function	Correctness-based (+1.0 correct, −1.0 incorrect)
LoRA rank / α / dropout	16 / 32 / 0.05
Precision	BF16
Seed	42

Table 15: QLoRA fine-tuning hyperparameters (shared across all model families and contamination conditions).

Hyperparameter	Value
Quantization	4-bit NF4
LoRA rank	16
LoRA α	32
LoRA dropout	0.05
Target modules	`q_proj`, `k_proj`, `v_proj`, `o_proj`
Learning rate	2×10^{-4}
Optimizer	AdamW
Warmup ratio	0.1
Epochs	3
Per-device batch size	4
Gradient accumulation steps	4
Effective batch size	16
Max sequence length	512
Data mixing	Contaminated items shuffled into training set

L Contamination Run Reproducibility

Item selection seed robustness. As a pipeline validation, we retrained Llama at 15% and 50% contamination using a different item selection seed (seed=42 vs. the original seed=999), which selects a distinct subset of MMLU items for contamination. The resulting ℓ_z^* values are numerically identical (−2.992 vs. −2.992 at 15%; −3.331 vs. −3.331 at 50%), confirming that ℓ_z^* is robust to which specific items are contaminated and depends primarily on the contamination proportion.

M Use of AI Assistants

The authors used AI-based writing assistants, including Claude, for limited support during manuscript preparation, including grammar polishing, wording suggestions, LaTeX editing, and code/debugging assistance. The tools were not used to generate the core research ideas, experimental design, methodological decisions, data analysis, or scientific conclusions. All AI-assisted text, code suggestions, and references were manually reviewed and edited by the authors.

Table 16: Reproducibility of contamination effect sizes across independent QLoRA runs (Qwen2.5-7B-Instruct, MMLU). Each row shows Cohen's $|d|$ from an independent training seed against the shared n=10 clean baseline ($\bar{\ell}_z^* = 8.68$, $s = 0.74$). All runs use identical hyperparameters; only the random seed differs.

	Free-θ $|d|$			Fixed-θ $|d|$		
Seed	15%	25%	50%	15%	25%	50%
42 (original)	6.57	8.22	1.90	4.77	4.96	14.52
123	5.09	8.78	5.42	7.79	5.37	12.64
456	4.41	5.31	7.91	7.99	7.34	9.80
Mean	5.36	7.44	5.08	6.85	5.89	12.32
SD	1.10	1.86	3.02	1.80	1.27	2.38
CV (%)	20.3	25.2	59.6	26.3	21.7	19.3

Note. Fixed-θ estimation anchors $\hat{\theta}$ at the clean-baseline mean (θ=-0.102), removing θ-shift absorption. The 50% free-θ column exhibits high CV (59.6%), reflecting seed-dependent θ-shift magnitude; fixed-θ reduces this to 19.3%, confirming that cross-seed variability at high contamination is driven by ability-estimate instability rather than true signal variance.

Table 17: Multi-seed Cohen's $|d|$ for HT and accuracy z-score statistics (Qwen2.5-7B-Instruct, MMLU), computed using the chi-squared ratio HT statistic and bootstrap Cohen's d (10,000 resamples). These supplement the ℓ_z^* values in Table 3.

	HT $|d|$			Acc. z $|d|$		
Seed	15%	25%	50%	15%	25%	50%
42 (original)	7.26	8.90	1.66	42.3	49.8	68.7
123	5.52	9.51	5.80	47.3	53.4	70.8
456	4.72	5.10	8.41	45.4	51.3	70.4
Mean	5.83	7.84	5.29	45.0	51.5	70.0
SD	1.30	2.39	3.41	2.5	1.8	1.1

Note. HT exhibits high cross-seed variance at 50% (SD=3.41), mirroring the ℓ_z^* pattern: the non-monotonicity observed in seed 42 ($|d|$=1.66) does not generalize across seeds. Accuracy z-scores are highly stable (CV $<$ 6%) and monotonically increasing with contamination level.

Table 18: Multi-seed Cohen's d for Llama-3.1-8B across all contamination levels (MMLU). Each cell shows Cohen's d from an independent training seed against the shared n=10 clean baseline ($\bar{\ell}_z^* = +0.26$, $s = 1.54$). The 5% condition exhibits higher cross-seed variance (SD=0.99), reflecting the weaker signal from $\sim$700 contaminated items (5%$\times$14,042).

	Per-seed d				
Contam. level	s_1	s_2	s_3	Mean	SD
5%	−3.22	−3.06	−1.44	−2.57	0.99
15%	−2.11	−3.21	−3.09	−2.80	0.60
25%	−3.16	−2.28	−2.24	−2.56	0.52
50%	−2.33	−2.59	−1.99	−2.30	0.30

Note. Unlike Qwen, Llama exhibits no non-monotonic dose-response: mean $|d|$ is approximately flat across contamination levels (2.30–2.80), consistent with its near-zero baseline ℓ_z^* (+0.26) and negligible θ-shift absorption (§4.2). Cross-seed variance decreases with contamination level (CV: 39% at 5% $\rightarrow$ 13% at 50%), suggesting that higher contamination produces a more stable signal despite the flat mean trajectory.

Table 19: Multi-seed Cohen's d for Mistral-7B-v0.3 across contamination levels (MMLU). Each cell shows Cohen's d from an independent training seed against the shared n=10 clean baseline ($\bar{\ell}_z^* = +4.02$, $s = 2.05$). The 5% condition has only n=2 seeds; its high SD reflects this limited sample.

	Per-seed d				
Contam. level	s_1	s_2	s_3	Mean	SD
5%	−2.82	−4.83	—	−3.83	1.42
15%	−2.64	−3.98	−3.48	−3.37	0.68
50%	−3.50	−4.25	−4.51	−4.09	0.53